

DAVIDE BALDELLI

PhD researcher at Polytechnique Montréal and Mila

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EDUCATION

MILA and Polytechnique Montréal

Jan 2025 - Current

PhD in Computer Engineering (GPA: 4.0)

- Supervisors: Sarath Chandar, Amal Zouaq.
- Courses included Representation Learning, Reinforcement Learning.

University of Bologna

Sept 2021 - Oct 2023

MSc in Artificial Intelligence (Grade: 110/110 cum laude)

- Courses included Programming for Data Science, Knowledge Representation and Reasoning, Statistical Theory and Methods, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing.
-  Thesis: A TWO-step LLM Augmented distillation method for passage Reranking.

University of Florence

Sept 2017 - Mar 2021

BSc in Mathematics (Grade: 110/110)

- Courses included Analysis, Linear Algebra, Geometry, Probability and Statistics, Logic and Computability.
-  Thesis: Big Deviations for Markov Chains.

PUBLICATIONS

ϕ TD: Distributional Reinforcement Learning using Characteristic Functions

NeurIPS 2026

Thomas Mousseau, Tyler Kastner, Davide Baldelli, Amir-massoud Farahmand

Probabilistic Calibration Is a Trainable Capability in Language Models

PrePrint

Davide Baldelli, Sruthi Kuriakose, Maryam Hashemzadeh, Amal Zouaq, Sarath Chandar

LLMs Can't Play Hangman: On the Necessity of a Private Working Memory for Language Agents

CoLLAs 2026 (Oral)

MemAgents Workshop @ ICLR 2026

Davide Baldelli, Ali Parviz, Amal Zouaq, Sarath Chandar

CADmium: Fine-Tuning Code Language Models for Text-Driven Sequential CAD Design

TMLR

Prashant Govindarajan, Davide Baldelli, Jay Pathak, Quentin Fournier, Sarath Chandar

TWOLAR: A TWO-Step LLM-Augmented Distillation Method for Passage Reranking

ECIR 2024 (Oral)

Davide Baldelli, Junfeng Jiang, Akiko Aizawa, Paolo Torrioni

POSITIONS AND PROGRAMS

MATS Fellowship (10.0 + 10.1 Extension)

Jun 2026 - Present

- Supervised by Mirko Bronzi and Damiano Fornasiere from LawZero.
- Investigating introspection capabilities in language models.
- Selected for the MATS 10.1 extension phase (from Sep 2026) to continue the project.

ARENA 7.0 (Alignment Research Engineer Accelerator)

Jan 2026

- Technical AI safety program focused on building ML engineering skills for alignment research.

Loop AI Labs Inc.

Sept 2022 - Dec 2024

- Machine Learning Engineer.
- Co-supervised a master's thesis titled "Self Assessment Tool for Medical Exam Candidates" in collaboration with the University of Padova.

National Institute of Informatics (NII)

Mar 2023 - Aug 2023

- International Internship Program.
- Supervisors: Akiko Aizawa, Paolo Torrioni.

- Developing an LLM based text-reranker.

SERVICE AND LEADERSHIP

INF8245AE Machine Learning – Polytechnique Montréal *Fall 2026*
Teaching Assistant

Fatima Al-Fihri Predoctoral Fellowship *Fall 2026*
Research Mentor

- Mentoring a fellow on a machine learning research project through the Fatima Institute for Global AI Research, a program supporting prospective PhD applicants from low- and middle-income countries.

SPAR (Supervised Program for Alignment Research) *Fall 2026*
Research Mentor

- Mentoring two research fellows on building robust activation monitors for language models.

CoLLAs 2026 (Conference on Lifelong Learning Agents) *Nov 2025 - Current*
Web Communication Chair

- Managing the official conference website and digital presence for the 2026 edition.

MILA *Summer 2025*
Teaching Assistant – Summer High School Internships

- Designed and delivered educational workshops on Machine Learning and Probability for high school students.

UILO – Career Orientation Platform *Sept 2024 – Present*
Co-Creator and Technical Lead

- Built UILO, a career orientation platform for post-secondary education with university and career databases and psychometry-based recommendations.

Belle Parole Arci APS *Nov 2023 - Current*
Co-Founder, Non-Profit Association

- Organized community events and communications, built the organization’s website belleparoleaps.it, and secured grant funding for organizational activities.

ADDITIONAL SKILLS, SERVICE, AND ACHIEVEMENTS

Reviewing	CoLLAs 2026; MemAgent Workshop at ICLR 2026.
Skills	PyTorch, Hugging Face, vLLM, LangChain/LangGraph, Python, Linux, L ^A T _E X, Weights & Biases, Figma.
Recognition	Selected Participant, Y Combinator AI Startup School (Jun 2025).
Languages	English (professional proficiency); Italian (native proficiency).